INT374 : DATA ANALYTICS WITH POWER BI

PROJECT REPORT

(Project Semester August 2026-January 2027)

BLINKIT SALES ANALYSIS



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CERTIFICATE

This is to certify that Utpal Kumar bearing Registration No. 62600440 has successfully completed the course project titled **"BLINKIT SALES ANALYSIS Using Power BI"** under my guidance and supervision for the course INT374. The work presented is original and meets the academic requirements.

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Date:24/10/2025

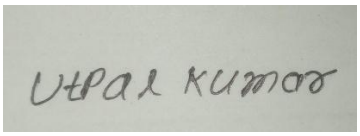
DECLARATION

I, Utpal Kumar, student of B.Tech Computer Science & Engineering under CSE Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine. This project work is original and has not been submitted elsewhere for any academic award.

Date: 24/10/2026

Registration No: 62600440

Signature:



Name: Utpal kumar

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my mentor Mr. Jaffar Amin Chacket , Assistant Professor at Lovely Professional University, for their continuous support, guidance, and encouragement throughout the course of this project.

I would also like to thank all faculty members of the School of Computer Science and Engineering for providing the resources and learning environment that enabled me to undertake this research. Lastly, I extend my heartfelt thanks to my family and friends for their moral support and motivation.

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1. INTRODUCTION

In today's competitive business environment, data analysis plays an important role in understanding sales performance, customer preferences, and business growth. Retail businesses generate large amounts of data every day through product sales, customer orders, outlet operations, and inventory management. Analyzing this data helps businesses identify trends, evaluate performance, and make better decisions.

The **Blinkit Sales Analysis Using Power BI** project focuses on analyzing retail sales data and presenting meaningful information through an interactive dashboard. Microsoft Power BI is used to transform raw data into visual reports that make it easier to understand sales performance, product categories, outlet characteristics, and other important business indicators.

The project uses a dataset obtained from Kaggle. The dataset is examined and prepared for analysis before being represented through suitable charts, graphs, tables, and key performance indicators (KPIs). These visualizations help users explore the available data and understand the relationship between different sales-related factors.

The dashboard is designed to provide a clear overview of sales performance. It can help analyze total sales, item performance, outlet-wise sales, product categories, and customer ratings wherever the corresponding fields are available in the dataset. Interactive filters and visual elements allow users to explore the data from different perspectives.

Power BI provides useful features such as Power Query for data transformation, data modeling for organizing information, and DAX (Data Analysis Expressions) for creating calculated measures. These features support the development of an organized and interactive analytical solution.

The main purpose of this project is to demonstrate how business intelligence and data visualization techniques can convert raw retail data into useful insights. The project also provides practical experience in data preparation, dashboard development, and analytical reporting.

Overall, this project demonstrates the application of Microsoft Power BI in retail sales analysis and shows how interactive dashboards can support better understanding of business performance and data-driven decision-making.

2. OBJECTIVES OF THE PROJECT

The main objective of the Blinkit Sales Analysis Using Power BI project is to analyze retail sales data and present meaningful insights through an interactive dashboard. The project aims to understand sales performance, product trends, and outlet-related factors using data visualization and business intelligence techniques.

The key objectives of the project are:

1. To Analyze Overall Sales Performance: To examine total sales and understand the overall performance of the retail business using the available dataset.
2. To Evaluate Product Performance: To identify product categories and individual items that contribute significantly to total sales.
3. To Analyze Outlet Performance: To compare sales across different outlet types, sizes, locations, and establishment years wherever the relevant data is available.
4. To Understand Customer Preferences: To analyze available customer ratings and other relevant attributes to understand customer satisfaction patterns.
5. To Identify Sales Trends: To examine variations in sales across product categories and outlet characteristics to identify useful patterns.
6. To Clean and Prepare Data: To use Power Query for handling missing values, checking duplicate records, correcting data types, and preparing the dataset for analysis where required.
7. To Develop an Interactive Dashboard: To create an informative and user-friendly Power BI dashboard using suitable charts, graphs, KPI cards, and filters.
8. To Apply Data Analysis Techniques: To use appropriate calculations, aggregations, and DAX measures to summarize and analyze sales data.
9. To Support Data-Driven Decision-Making: To present information in a visual format that helps users understand business performance and identify areas for improvement.
10. To Gain Practical Experience: To develop practical skills in data analytics, data visualization, report preparation, and business intelligence using Microsoft Power BI.

Through these objectives, the project aims to demonstrate how Power BI can be used to convert raw retail sales data into understandable visual insights and support effective business analysis.

3. DATASET OVERVIEW

The dataset used in this project, *Blinkit Sales Analysis Using Power BI*, was obtained from Kaggle, an online platform that provides datasets for data science, data analytics, and machine learning projects. The dataset serves as the primary source of information for analyzing retail sales performance and developing an interactive Power BI dashboard.

The dataset contains retail-related information that can be used to study product sales, item categories, outlet characteristics, and other business-related factors. By analyzing these attributes, the project aims to understand the distribution of sales and identify patterns that may help in evaluating business performance.

The dataset is imported into Microsoft Power BI, where it can be examined, transformed, and visualized. Data preparation is an important step because the quality and consistency of the data directly affect the reliability of the analysis.

3.1 Source of the Dataset

The dataset was collected from Kaggle and used for academic and analytical purposes. It provides the input data required to create visualizations and perform sales analysis in Power BI.

Dataset Source: Kaggle

Dataset Link: Insert the exact Kaggle dataset URL used in this project.

3.2 Purpose of the Dataset

The primary purpose of the dataset is to support the analysis of retail sales information. It helps organize business data into meaningful categories so that comparisons and summaries can be created using Power BI.

The dataset is used to:

- Analyze overall sales performance.
- Compare sales across available product categories.
- Examine item-level sales contributions.
- Study outlet-related sales patterns.
- Analyze customer ratings, if available.
- **Create interactive charts, graphs, and key performance indicators.**

3.3 Dataset Structure

The dataset is organized into rows and columns. Each row represents a record, while each column represents an attribute associated with that record. The exact number of records, columns, and field names depends on the dataset version used in the project.

The following types of information may be available in a typical Blinkit sales dataset:

- Item Information: Product identifiers, item names, or item categories.
- Sales Information: Item sales values and other available sales measures.
- Product Attributes: Product characteristics, such as fat content, if included.

- Outlet Information: Outlet identifiers, outlet types, sizes, and locations, if included.
- Customer Feedback: Ratings or related feedback fields, if included.

These attributes can be used to summarize and compare sales data through suitable visualizations.

3.4 Importance of the Dataset

The dataset is important because it provides a structured foundation for the entire project. Instead of examining individual records manually, Power BI can summarize the information and present it through interactive visualizations.

The dataset also allows users to explore different aspects of sales performance, identify patterns in the available information, and understand how data analytics can support retail business decisions.

Overall, the dataset forms the foundation of the Blinkit Sales Analysis project and enables the transformation of raw data into meaningful business insights.

4. FEATURE DESCRIPTION

The feature description section explains the important attributes available in the Blinkit sales dataset. These features help organize the data and perform different types of sales analysis using Microsoft Power BI. Each feature represents a particular characteristic of a product, sales transaction, or outlet.

The following features are commonly available in Blinkit sales datasets. The final list should be matched with the actual columns present in the dataset used for this project.

4.1 Item Identifier

This feature represents the unique identification code assigned to a product. It helps distinguish individual items and organize product-related information.

4.2 Item Type

This feature represents the category to which a product belongs. It can be used to compare sales performance across different product categories.

4.3 Item Fat Content

This feature indicates the fat-content classification of an item, where applicable. It can be used to compare sales across different fat-content categories.

4.4 Item Visibility

This feature represents the visibility of a product within an outlet, if included in the dataset. It can help examine the relationship between product visibility and sales performance.

4.5 Item Weight

This feature represents the weight of a product. It can be used to study product characteristics and compare sales across different product weights.

4.6 Item Outlet Sales

This feature represents the sales value associated with an item at an outlet. It is an important numerical feature for calculating total sales and comparing product performance. In this project, dashboard sales values are displayed in US dollars ($), where applicable.

4.7 Outlet Identifier

This feature identifies a particular retail outlet. It helps group and compare sales information across outlets.

4.8 Outlet Establishment Year

This feature indicates the year in which an outlet was established, if available. It can help examine sales patterns based on outlet establishment year.

4.9 Outlet Size

This feature represents the size classification of an outlet, such as small, medium, or high-capacity categories, depending on the dataset. It can be used to compare sales across outlet sizes.

4.10 Outlet Location Type

This feature describes the location classification of an outlet, if included. It helps analyze sales distribution across different outlet locations.

4.11 Outlet Type

This feature identifies the category or format of an outlet. It can be used to compare sales performance across different outlet types.

4.12 Customer Rating

This feature represents the customer rating associated with an item or outlet, if available. It can be used to summarize customer feedback and examine rating patterns.

Summary

These features provide the foundation for analyzing sales performance, comparing product categories, and evaluating outlet-related patterns. In Power BI, the appropriate fields can be used in charts, tables, filters, and key performance indicators to make the analysis easier to understand.

The actual feature list and descriptions must correspond to the columns present in the dataset used for this project.

5. TOOLS AND TECHNOLOGIES USED

The Blinkit Sales Analysis project uses data analytics and business intelligence tools to transform raw retail sales data into meaningful visual insights. The following tools and technologies support the preparation, analysis, and visualization of the dataset.

5.1 Microsoft Power BI Desktop

Microsoft Power BI Desktop is the primary tool used to develop the Blinkit Sales Analysis dashboard. It provides features for importing data, creating visualizations, building reports, and presenting business information through an interactive interface.

Power BI is used to represent sales performance through suitable charts, tables, and Key Performance Indicator (KPI) cards. It helps users compare product categories and outlet-related information based on the fields available in the dataset.

5.2 Power Query

Power Query is a data transformation tool available in Power BI. It helps prepare raw data for analysis by providing options to remove unnecessary columns, handle missing values, identify duplicate records, correct data types, and standardize inconsistent values.

In this project, Power Query can be used to prepare the Blinkit dataset before creating the dashboard. These transformations help improve data consistency and support reliable analysis.

5.3 Data Analysis Expressions (DAX)

Data Analysis Expressions (DAX) is a formula language used in Power BI to create calculated columns, measures, and analytical calculations.

DAX can be used to calculate important sales indicators, such as total sales, average sales, item counts, and other relevant metrics, depending on the dataset and project requirements. Measures also help dashboard visuals respond dynamically to filters and slicers.

5.4 Data Visualization

Data visualization is the process of presenting information through graphical representations. Power BI provides various visualization options, including bar charts, column charts, line charts, donut charts, tables, and KPI cards.

In this project, appropriate visuals are used to present sales information clearly and make comparisons easier. The selection of each visual depends on the type of data and the analytical objective.

5.5 Kaggle Dataset

Kaggle is an online platform that provides datasets for data analysis, machine learning, and research projects. The Blinkit sales dataset used in this project was obtained from Kaggle.

The dataset provides the input information required for sales analysis and dashboard development. Its actual fields and structure determine the calculations and visualizations that can be created.

5.6 Data Modeling

Data modeling involves organizing data and establishing relationships between tables when multiple tables are used. In Power BI, a suitable data model helps maintain consistency and supports accurate calculations.

The data model for this project depends on the structure of the imported dataset. If the project uses a single table, the analysis can be performed directly on that table without requiring additional table relationships.

Summary

The combination of Power BI Desktop, Power Query, DAX, data visualization techniques, and the Kaggle dataset provides the foundation for the Blinkit Sales Analysis project. These tools support data preparation, analytical calculations, and the creation of an interactive dashboard for understanding retail sales performance.

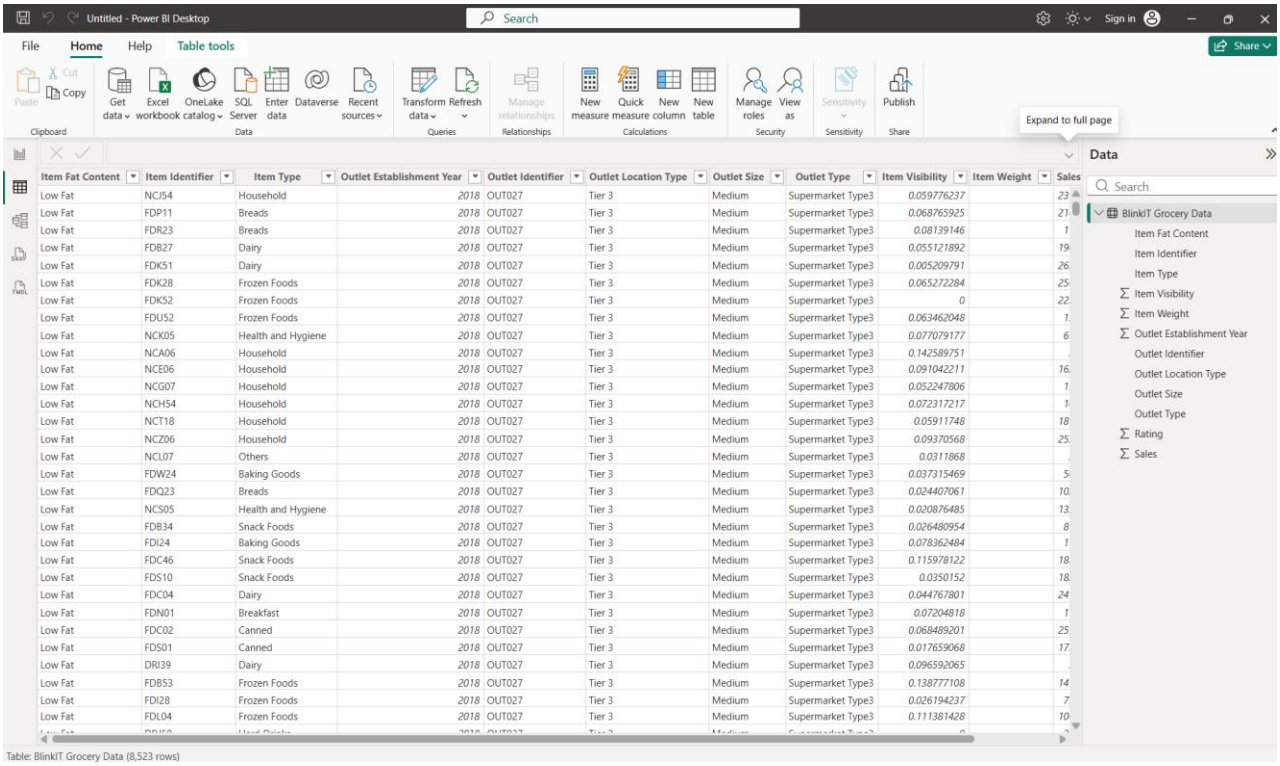
6. DATA PREPARATION AND CLEANING

Data preparation and cleaning is an important step in the Blinkit Sales Analysis project. Before creating visualizations and analyzing sales performance in Microsoft Power BI, the dataset needs to be checked for missing values, duplicate records, inconsistent entries, and incorrect data types.

Microsoft Power Query can be used to prepare the dataset and make it suitable for analysis.

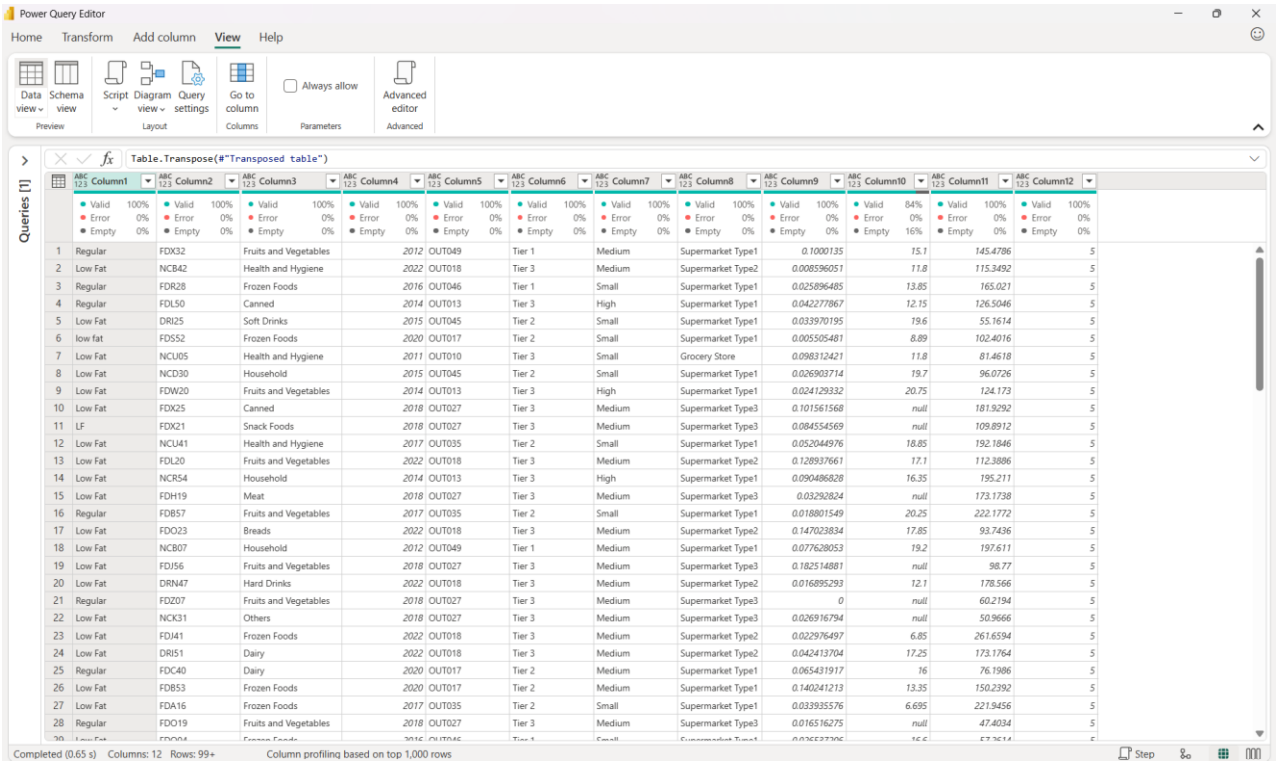
6.1 Data Import

The Blinkit sales dataset was obtained from Kaggle and imported into Microsoft Power BI Desktop. After importing the dataset, the available columns and data types were examined to understand the structure of the data.



6.2 Checking Missing Values

The dataset was checked for blank or missing values in important columns. Missing values can affect calculations and produce misleading results. Any missing values, if present, should be handled according to the nature of the column and the requirements of the analysis.



6.3 Removing Duplicate Records

Duplicate records can affect total sales and other calculations. The dataset was checked for duplicate rows. Duplicate records, if identified, should be removed only after confirming that they are unnecessary repeated entries.

6.4 Correcting Data Types

The data types of the columns should be verified to ensure accurate calculations. Numerical fields, such as sales and item weight, should use appropriate numeric data types. Text fields, such as item type and outlet type, should be stored as text. Date-related fields should use suitable date or whole-number formats according to the dataset.

6.5 Handling Inconsistent Values

Categorical columns were reviewed for inconsistent labels, spelling differences, or variations in capitalization. Where required, these values should be standardized to ensure that similar categories are grouped correctly in the dashboard.

6.6 Preparing Data for Analysis

After the required cleaning and formatting steps, the dataset becomes more suitable for analysis. The prepared data can then be used to create charts, tables, Key Performance Indicators (KPIs), and interactive dashboard elements in Power BI.

6.7 Importance of Data Cleaning

Data cleaning improves data quality and helps reduce errors in calculations and visualizations. It also supports reliable comparison of sales across product categories and outlet types.

Therefore, data preparation and cleaning form the foundation of the Blinkit Sales Analysis dashboard and help ensure that the insights presented are based on properly organized data.

7. ANALYSIS ON DATASET

The Blinkit sales dataset is analyzed using Microsoft Power BI to understand sales performance across different products and outlets.

The analysis focuses on the following areas:

1. **Sales Analysis:** Understanding overall sales performance.



2. **Product Analysis:** Comparing sales across different item types.



3. **Outlet Analysis:** Analyzing sales based on outlet type, size, and location.
4. **Item Fat Content Analysis:** Comparing sales of low-fat and regular products.
5. **Customer Ratings:** Examining customer ratings, if available in the dataset.

Power BI charts, tables, and KPI cards help present the results in an interactive and easy-to-understand format. This analysis supports better understanding of product performance and outlet sales.

8. DASHBOARD DESIGN AND INTERACTIVITY

The Power BI dashboard developed in this project was designed with a strong emphasis on clarity, usability, and professional presentation. Since the target users of the dashboard include planners, administrators, and policymakers who may not have a technical background in data analytics, the design prioritizes simplicity and ease of interpretation. The primary objective of the dashboard is to transform complex crime data into an intuitive visual format that supports quick understanding and informed decision-making.

Special attention was given to ensuring that the dashboard provides both high-level summaries and detailed insights. Key performance indicators (KPIs) are prominently displayed to give users an immediate overview of physical and financial progress, while detailed charts and visuals allow deeper exploration of the data. This layered design approach ensures that users can access the required level of detail without being overwhelmed by information.

8.1 Use of Interactive Slicers and Filters

To enhance interactivity, slicers were provided for important dimensions such as year, area, crime type, and victim characteristics. These slicers allow users to dynamically filter the crime data and view patterns for a specific time period, location, or crime category. For example, a user can select a particular year to instantly view total crimes, weapon-related crimes, victim distribution, and clearance status for that year.

As slicer selections are changed, all visuals on the dashboard update automatically. This dynamic behavior ensures a smooth and responsive user experience, allowing users to explore multiple crime scenarios without refreshing or reloading the dashboard. Such interactivity makes the dashboard suitable for exploratory crime analysis and comparative evaluation across different years and regions.

8.2 Drill-Down and Hierarchical Navigation

The dashboard also supports drill-down functionality, which enables users to move from a summarized view of crime data to more detailed levels of analysis. For instance, users can begin with an overall crime overview and then drill down to analyze crime trends by year, area, or crime category.

Drill-down features are particularly useful in identifying problem areas. If a particular region shows a high number of weapon-related crimes or low clearance rates, users can drill down further to analyze victim demographics, weapon types, or monthly trends. This feature supports deeper investigation without cluttering the dashboard with excessive visuals.

8.3 Cross-Filtering and Cross-Highlighting

To further improve analytical capability, cross-filtering and cross-highlighting features were enabled across all visuals. This means that selecting a data point in one chart automatically highlights related data in other visuals. For example, selecting a specific area in a bar chart will instantly update KPI cards, maps, and trend charts to reflect crime patterns for that area.

This interconnected behavior enhances the exploratory nature of the dashboard and helps users identify relationships between crime trends, weapon involvement, and victim demographics. It also ensures consistency across visuals and improves overall interpretation of crime data.

8.4 Visual Design and Theme Selection

A dark-themed design was chosen for the dashboard to enhance readability and provide a professional appearance. Dark backgrounds help reduce visual strain, especially when dashboards are viewed for extended periods. Contrasting colors were carefully selected so that important KPIs, charts, and values stand out clearly against the background.

Consistent color usage was maintained across similar crime metrics to avoid confusion. For example, weapon-related crime metrics follow a consistent color pattern, while victim-related and clearance-related metrics use different but harmonized shades. Font style, size, alignment, and

spacing were kept uniform across all dashboard pages. Clear titles, axis labels, and legends ensure that each visual is self-explanatory and easy to understand.

8.5 Selection of Visuals

The dashboard uses a combination of KPI cards, bar charts, line charts, maps, donut charts, treemaps, and forecast visuals. Each visual was selected based on its suitability for representing specific crime-related insights.

KPI cards provide a quick snapshot of total crimes, weapon crimes, and clearance-related indicators. Bar and column charts enable comparison of crime counts across areas, weapon types, and victim categories. Line charts support trend and forecast analysis of crime incidents over time.

Maps highlight geographical concentration of crimes, while donut and treemap visuals show proportional contribution of crime types and weapon categories.

This thoughtful selection of visuals ensures that crime data is presented in the most effective and meaningful way.

8.6 User Experience and Decision Support

From a user experience perspective, the dashboard was designed to minimize complexity while maximizing analytical insight. Users do not require prior technical knowledge of Power BI to navigate the dashboard. Intuitive slicers, clear navigation buttons, and interactive visuals guide users naturally through the analysis.

Overall, the dashboard functions as an effective decision-support tool. It enables stakeholders to quickly identify crime hotspots, monitor weapon-related crime trends, analyze victim vulnerability, and evaluate clearance performance. By presenting multiple crime indicators in an integrated and interactive manner, the dashboard supports proactive planning, improved monitoring, and data-driven public safety decisions.

In conclusion, dashboard design and interactivity play a crucial role in the success of this project. By combining strong visual design principles with powerful interactive features, the Power BI dashboard effectively transforms crime data into actionable insights that support crime prevention and public safety management.

9. KEY INSIGHTS AND FINDINGS

The Blinkit Sales Analysis dashboard provides useful insights into sales performance, product categories, and outlet operations. Microsoft Power BI helps present the data through charts, graphs, KPI cards, and interactive filters.

9.1 Sales Performance

The dashboard helps analyze overall sales and compare sales performance across different products and outlets. This makes it easier to identify categories and outlets that contribute significantly to total sales.

9.2 Product Category Analysis

Different item categories can be compared to understand their contribution to total sales. This analysis helps identify popular product categories and supports better product planning.

9.3 Outlet Performance

The dashboard allows comparison of outlets based on outlet type, size, and location. This helps users understand differences in sales performance across various outlet categories.

9.4 Item Fat Content Analysis

Sales can be compared between low-fat and regular products to understand their contribution to overall sales. This information can help analyze customer purchasing patterns.

9.5 Importance of the Findings

The dashboard presents sales information in a clear and interactive format. It helps users identify sales patterns, compare performance, and make informed business decisions.

Overall, the analysis demonstrates how Power BI can convert raw sales data into meaningful visual insights. Exact findings depend on the values and results shown in the completed dashboard.

10. CONCLUSION

The Blinkit Sales Analysis project demonstrates the use of Microsoft Power BI for analyzing and visualizing retail sales data. The project focuses on understanding sales performance, comparing product categories, and analyzing outlet-related information through interactive dashboards.

During this project, the dataset was studied and prepared for analysis. Different visualizations, such as charts, graphs, and KPI cards, were used to present important sales information in a clear and understandable manner. Power BI features help users explore the data and compare performance across different categories.

The dashboard provides a convenient way to understand sales patterns and identify areas that may require further analysis. It also demonstrates how data visualization tools can transform raw data into meaningful business information.

Overall, this project helped develop practical knowledge of Power BI, data visualization, data analysis, and dashboard design. It also highlights the importance of data-driven decision-making in the retail industry.

11. FUTURE SCOPE

The Blinkit Sales Analysis project can be improved further by adding advanced analytical features and more detailed visualizations. The current dashboard provides a foundation for understanding sales performance, product categories, and outlet operations.

The future scope of this project includes the following improvements:

1. Sales Forecasting: Predicting future sales using historical sales data and forecasting techniques.

2. Real-Time Data Analysis: Connecting the dashboard to regularly updated data sources to monitor sales more frequently.

3. Advanced Customer Analysis: Analyzing customer preferences and purchasing patterns if detailed customer data becomes available.

4. Profit Analysis: Including cost, profit, and profit-margin information to understand business profitability.

5. Improved Dashboard Features: Adding more filters, drill-through pages, and interactive visualizations for detailed analysis.

6. Automated Reporting: Setting up scheduled data refreshes and report sharing through Power BI Service, where supported.

These improvements could make the dashboard more useful for business planning and decision-making. With additional data and advanced analytical methods, the project could provide deeper insights into retail sales performance.

12. REFERENCES

The following resources were used as references for the Blinkit Sales Analysis project:

1. **Kaggle:** Used as the source of the Blinkit sales dataset.
   Website: https://www.kaggle.com/

2. **Microsoft Power BI:** Used for data visualization, sales analysis, and dashboard development.
   Website: https://powerbi.microsoft.com/

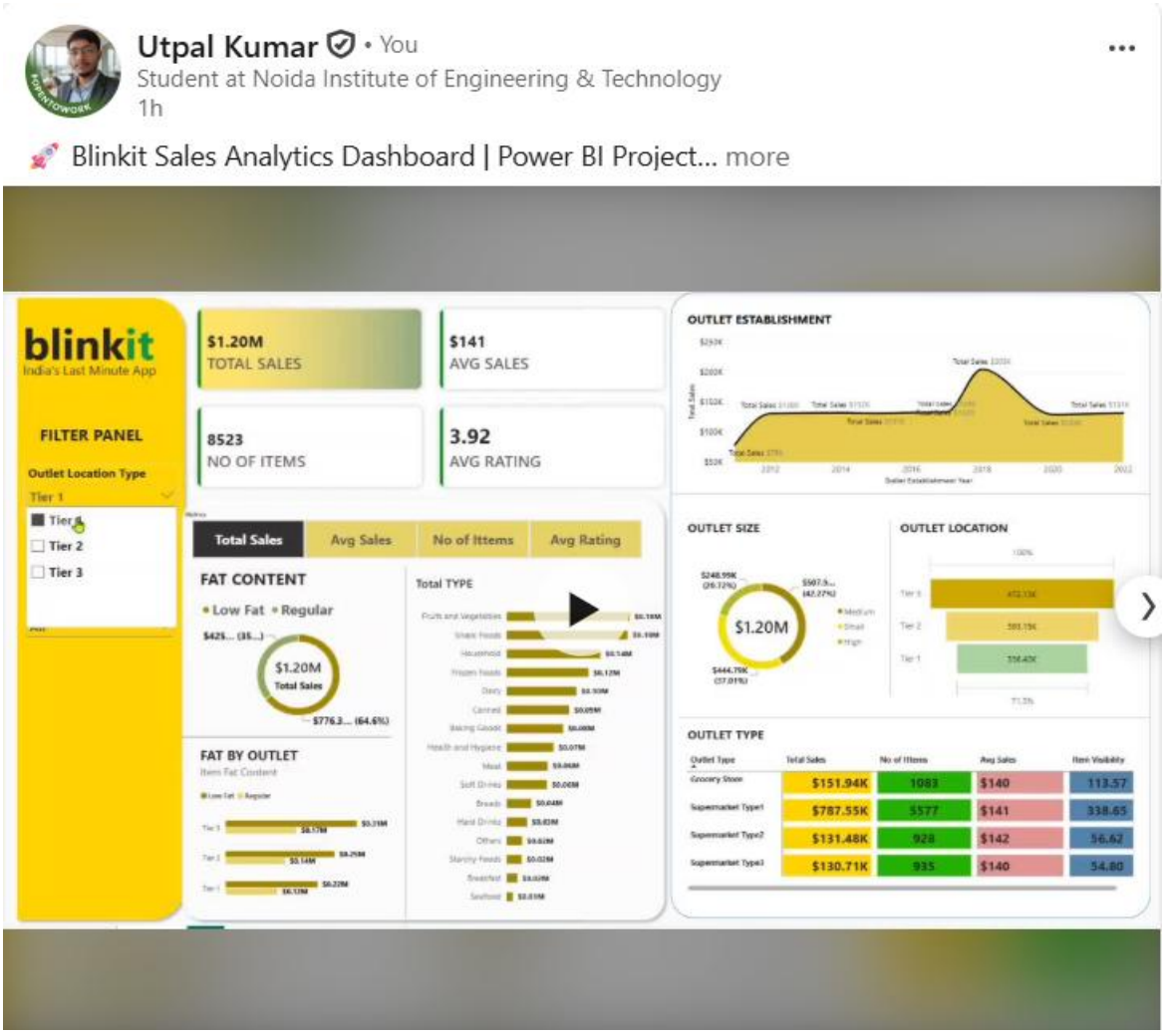
These resources provided useful information about the dataset, data preparation, visualization techniques, and dashboard development.

13. LINKEDIN

Link:

https://lnkd.in/p/dMHA8w-A

Screenshot:



14. DRIVE LINK

To view the PowerBI Dashboard File and Dataset :

https://drive.google.com/drive/folders/1CTxtrcogOOUjLyvTutTVhASqF0TMA5Tv?usp=drive_link